The Changing Life Cycle of Billboard Hits in the TikTok-Streaming Era

DS 112 Final Project | Abigail Chee and Troy Helenihi | June 2026

Abstract

The emergence of streaming platforms and short-form social media has reshaped how music reaches mainstream audiences. This study investigates how these changes are reflected in the life cycle and audio characteristics of successful Billboard Hot 100 songs. By combining Billboard chart data, Apple Music metadata, and computational audio analysis, we examined changes in chart performance, musical characteristics, and audio archetypes across the streaming and TikTok eras. We found that modern hit songs generally reach peak popularity more quickly, spend less time on the charts, and exhibit measurable shifts in dominant audio profiles over time. These findings indicate that the characteristics associated with mainstream success have evolved alongside changes in music discovery and consumption.

Research Question

Background / Motivation

Streaming platforms and short-form social media have fundamentally changed how audiences discover and consume music. While Billboard rankings have historically reflected sustained popularity, modern recommendation systems may reward rapid engagement and viral exposure. Despite these shifts, it remains unclear how the life cycle and audio characteristics of successful songs have evolved during the streaming and TikTok eras.

Research Question

How have the streaming and TikTok eras influenced the life cycle and audio characteristics of successful Billboard Hot 100 songs?

Key Questions

- Do modern hit songs reach peak popularity more quickly than earlier hits?
- Has chart longevity changed in the streaming and TikTok eras?
- Have the dominant audio profiles and archetypes of successful songs shifted over time?

BILLBOARD HIT LIFE CYCLE ANALYSIS

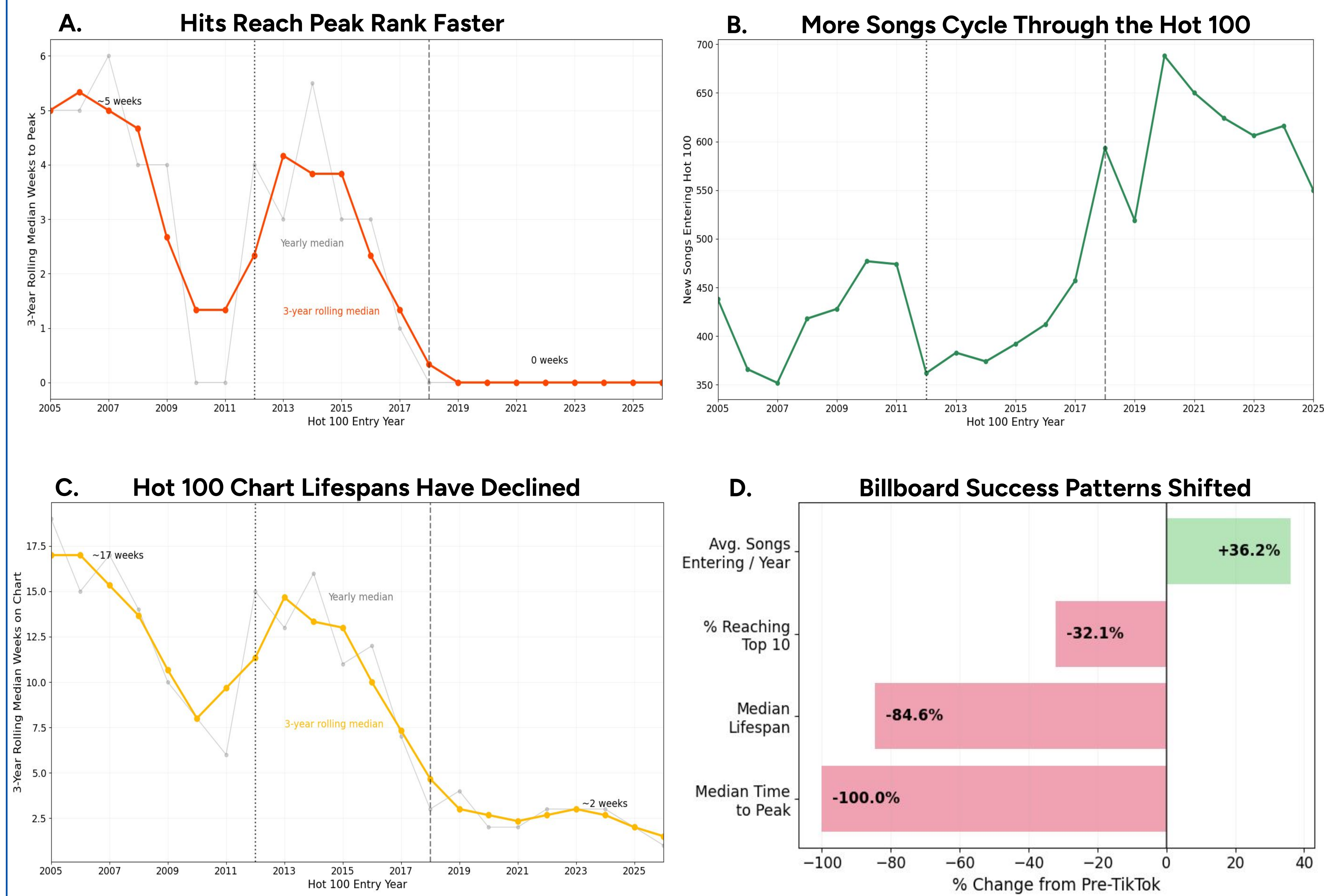


Figure 1. **Billboard success patterns shifted substantially in the TikTok-streaming era.** **(A)** Median time-to-peak declined from approximately 5 weeks in the mid-2000s to nearly 0 weeks after 2018, indicating that successful songs now reach peak popularity almost immediately after entering the chart. **(B)** The number of unique songs entering the Hot 100 each year increased substantially following the adoption of streaming and short-form social media, reflecting higher chart turnover. **(C)** Median chart lifespan decreased from roughly 17 weeks to approximately 2 weeks, suggesting that songs spend less time on the chart before being replaced by new entrants. **(D)** A pre- versus post-TikTok comparison summarizes these shifts, showing a 100% reduction in median time-to-peak, an 84.6% reduction in median lifespan, a 32.1% decline in the proportion of songs reaching the Top 10, and a 36.2% increase in the number of unique songs entering the Hot 100 annually. Together, these findings indicate that modern chart success is characterized by faster popularity cycles, shorter chart persistence, and increased turnover.

SONG AUDIO FEATURE ANALYSIS

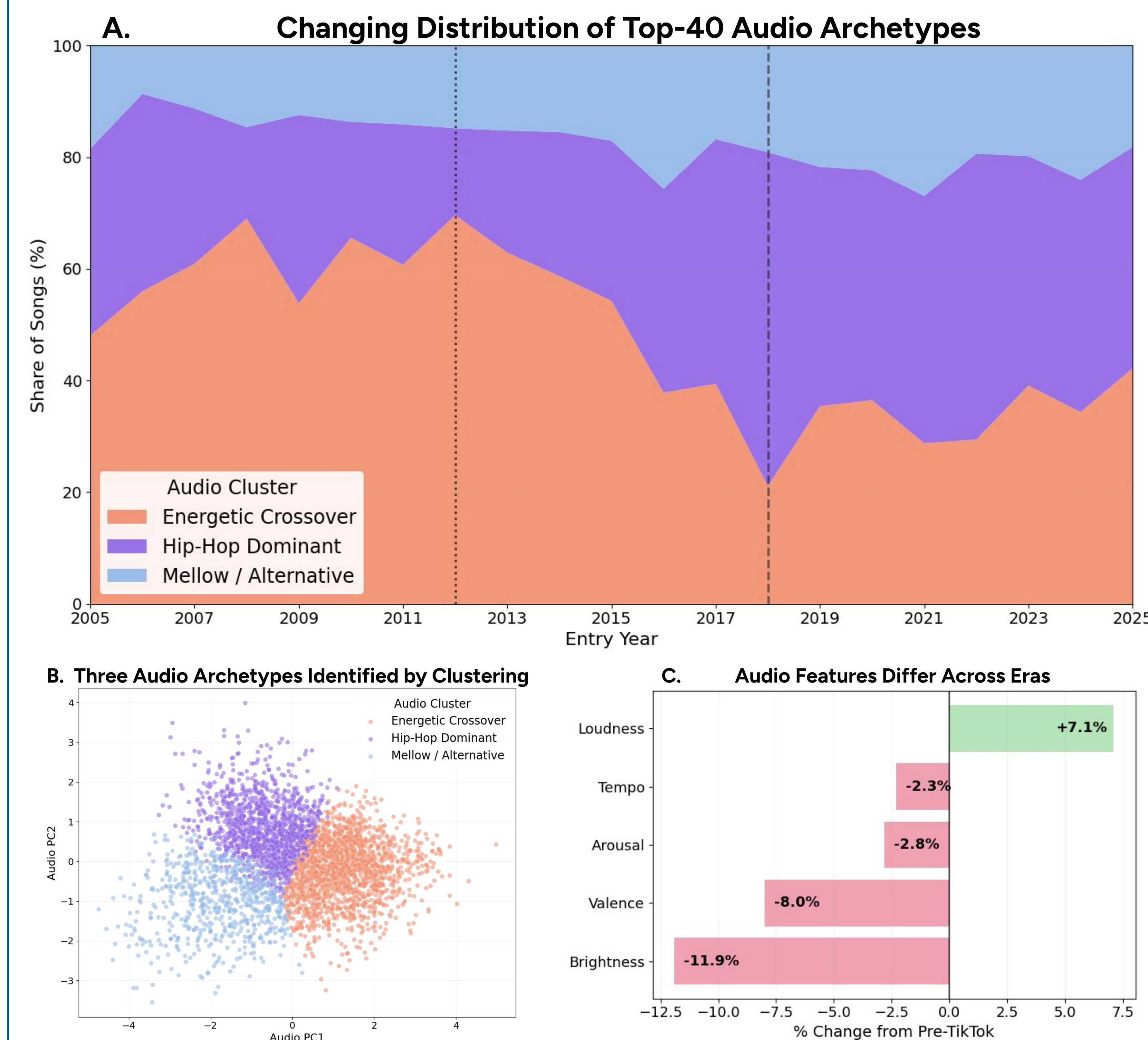
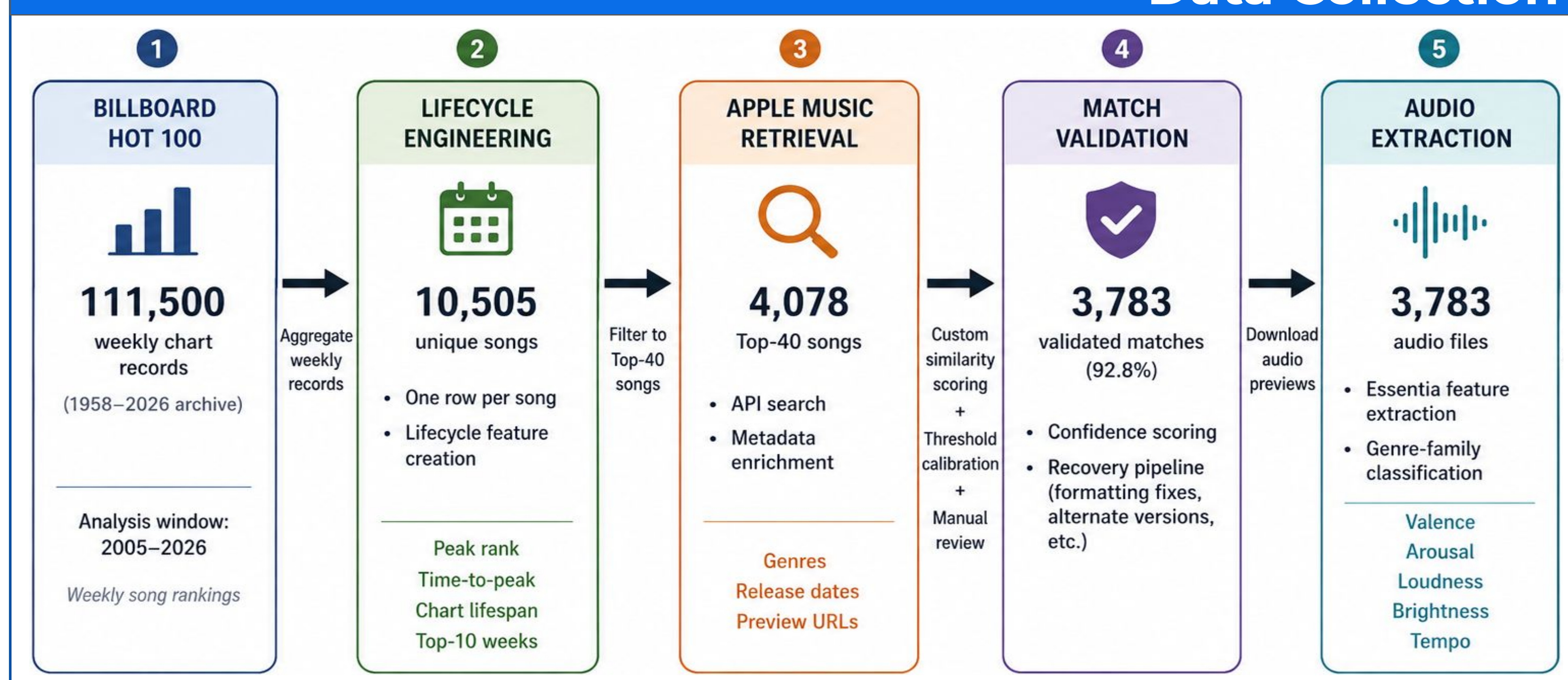


Figure 2. **Audio characteristics associated with successful songs evolved across the streaming and TikTok eras.** **(A)** The distribution of three audio archetypes shifted over time among Top-40 songs. Hip-Hop Dominant songs became increasingly prevalent after 2018, while Energetic Crossover songs declined in relative share. **(B)** K-means clustering identified three recurring audio archetypes—Energetic Crossover, Hip-Hop Dominant, and Mellow / Alternative. Principal component analysis (PCA) was used to visualize the cluster structure in two dimensions. **(C)** Comparison of pre- and post-TikTok eras revealed lower brightness (-11.9%), valence (-8.0%), tempo (-2.3%), and arousal (-2.8%), alongside increased loudness (+7.1%) among successful songs. Together, these findings suggest that changes in music discovery and consumption may influence the types of songs most likely to succeed.

Data Collection



Billboard Lifecycle Engineering

- Aggregated 111,500 weekly chart records into 10,505 unique songs
- Engineered lifecycle metrics including chart lifespan, time-to-peak, and Top-10 persistence
- Filtered to Top-40 songs for audio enrichment

Apple Music Matching

- Cleaned song and artist text to improve API retrieval
- Developed similarity-scoring system using title and artist agreement
- Reviewed low-confidence matches and recovered formatting mismatches

Audio Feature Extraction

- Downloaded 3,783 audio previews through Apple Music
- Extracted valence, arousal, loudness, brightness, and tempo using Essentia
- Generated audio archetypes through clustering analysis

Conclusions

- Successful songs now reach peak popularity much faster, often peaking immediately upon entering the Hot 100.
- Chart lifespans have declined substantially, indicating that songs are replaced more quickly than in previous eras.
- More unique songs enter the Hot 100 each year, suggesting increased turnover and a faster-moving competitive landscape.
- The distribution of successful audio archetypes and key audio characteristics has shifted across the streaming and TikTok eras.
- Overall, modern chart success is characterized by faster discovery, shorter life cycles, and changing patterns of musical popularity.

Limitations

- Billboard methodology has evolved over time, including changes in chart calculation and new consumption metrics.
- Billboard Hot 100 data reflects mainstream chart success and may not capture listening behavior occurring outside the chart ecosystem.
- Audio features were extracted from Apple Music preview clips rather than full-length recordings.
- Audio features capture sonic characteristics but do not account for lyrical content, cultural context, marketing, or artist popularity.
- The influence of streaming and TikTok is inferred through era-based comparisons rather than direct platform-level consumption data.